

Interpretable Text Embeddings and Text Similarity Explanation

A survey in EMNLP'25

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Interpretable Embeddings

❖ Text embeddings and similarities that humans can interpret (interpretable models)

Space-Shaping

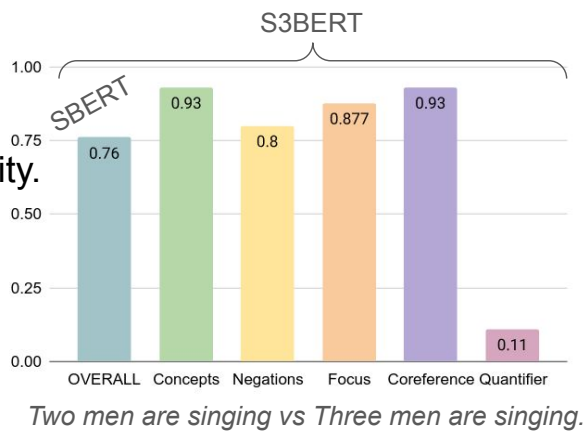
Explicitly encode specific aspects into embeddings

🧩 QA Features

Method: Frame embedding learning as answering predefined questions about a text (e.g., “Is it about sports?”).
Interpretability: Each dimension reflects a question-based feature derived from LLM-generated Yes/No supervision.
Outcome: Transparent embeddings that outperform baselines—defining a general question set remains challenging.

🧩 Sub-Embedding Features

Method: Decompose embeddings into *semantic subspaces* (focus, negation, etc.).
Example: S3BERT aligns subspaces with AMR-graph metrics for interpretable similarity.
Benefit: Enables fine-grained analysis of similarity across semantic dimensions.



⚓ Anchor Features

Method: Embedding dimensions express similarities to interpretable *prototype texts*.
Interpretability: Explanations arise from similarity to these anchors.

⚖️ Challenges & Opportunities

Trade-offs: Interpretability constraints may reduce general accuracy but can improve robustness and transparency.
Outlook: While reliant on predefined aspects, space shaping enables *custom, task-aligned embedding spaces*.

Set-Based

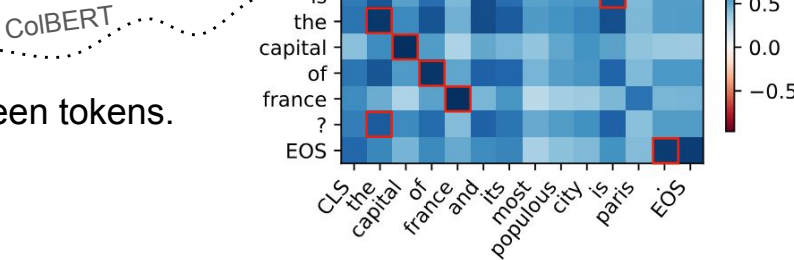
Produce multiple embeddings with dedicated meanings

🏷️ Token Weight Embeddings

Method: Aggregate token embeddings with learned importance weights derived from transformer layers, attention signals, or static token statistics.
Interpretability: Token-level contributions can be traced directly to the final text embedding.
Examples: SBERT-WK, Token-Attention, Model2Vec.

🔄 Sequential (Late-Interaction) Embeddings

Method: Compare token embeddings before pooling; **max-alignments** between tokens.
Interpretability: Alignment matrices highlight local semantic matches.



👁️ Multi-view Interpretation

Method: Generate multiple embeddings per text to capture alternative views.
Interpretability: Each embedding reflects a distinct aspect or interpretation of a text.

⚖️ Challenges & Opportunities

Set-based methods reveal detailed alignment and asymmetry but require more memory and computation. Sequential and multi-view variants scale with input size and may need multiple forward passes.

Sparsity

Achieve interpretability through sparse embeddings

⚙️ Unsupervised Sparsification

Method: Induce sparsity by reconstructing dense embeddings from sparse latent variables.
Finding: Some dimensions align with human-interpretable traits (e.g., spatial relations) but at the cost of reconstruction quality and task performance.

📄 Sparse Lexical Embeddings

Method: Represent texts as sparse, vocabulary-based term-weight vectors derived from transformer logits.
Benefit: Directly interpretable word-level features, including *expansion terms* beyond the input.
Use: Well suited for retrieval with efficient inverted indices.
Examples: DeepCT, SPARTA, SPLADE, and hybrid sparse–dense models.

⚖️ Challenges & Opportunities

Interpretability: Sparse lexical spaces are easily understood; unsupervised sparse ones less so.
Efficiency: High dimensionality demands specialized infrastructure.
Outlook: Strong potential for *transparent, retrieval-focused embeddings*.

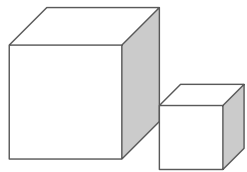


Structured Objects

Generate structured embeddings with interpretable relations

📦 Box Embeddings

Method: Represent texts as high-dimensional boxes; overlap measures similarity, containment captures entailment or asymmetry.
Interpretability: Geometry directly encodes logical relations (e.g., entailment).
Example: Soft probabilistic overlap (Gumbel-based) avoids high-dimensional decay.



🎲 Distributional Embeddings

Method: Model each text as a Gaussian distribution to represent multiple interpretations and uncertainty.
Interpretability: Mean captures central meaning; covariance encodes ambiguity.

⚙️ Operator Learning

Method: Train neural operators for semantic composition (e.g., union, fusion) within the embedding space.
Interpretability: Enables controlled manipulation of meaning; improves compositional reasoning.

⚖️ Challenges & Opportunities

Naturally captures asymmetric and relational meaning, but increased geometric complexity limits scalability to general tasks.

Post-hoc Explanation

❖ Explain black-box embeddings via additional computation or modeling

Attribution

Assign importance values to features and their interactions

⚙️ Integrated Jacobians

Method: Extend Integrated Gradients to dual encoder models, producing token-token importance matrices.

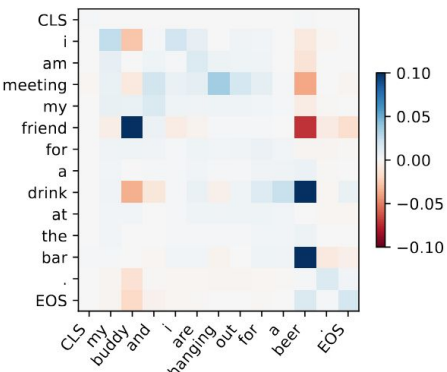
⚙️ Relevance Propagation

Method: Propagate relevance scores layer-wise through the model (BiLRP) for Siamese similarity architectures.

Interpretability: Both methods show how a dense text embedding model compares two texts on a token level.

⚖️ Challenges & Opportunities

Applicable to any differentiable embedding model, offering theoretical grounding but can be computationally expensive.



Global Explanation

Evaluate embedding relations of inputs with known properties

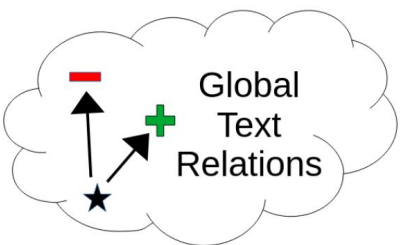
⚙️ Text Relations Analysis

Method: Construct evaluation sentence sets with known linguistic relations (e.g., negation, passivation, syntactic changes) and measures whether embeddings preserve these relations consistently.

Interpretability: on a global dataset level and limited to pre-defined properties.

⚖️ Challenges & Opportunities

Provides insights into embedding geometry and relation consistency. Limited by the need for curated datasets and property coverage, and requires significant manual effort.



Surrogate Modeling

Secondary interpretable models

⚙️ Interpretable Approximation

Method: Fit regression models on interpretable features to match similarity scores of embeddings.

Finding: Reveal linguistic features driving similarity.

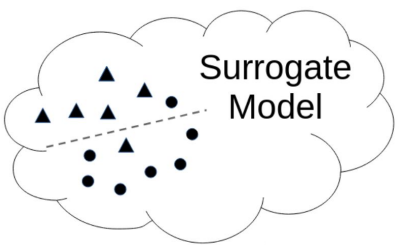
⚙️ Probing

Method: Train classifiers on top of embeddings to test if specific properties are linearly separable.

Finding: Probing uncovers syntactic, semantic, and surface-level information and is widely used for model analysis (e.g., MTEB).

⚖️ Challenges & Opportunities

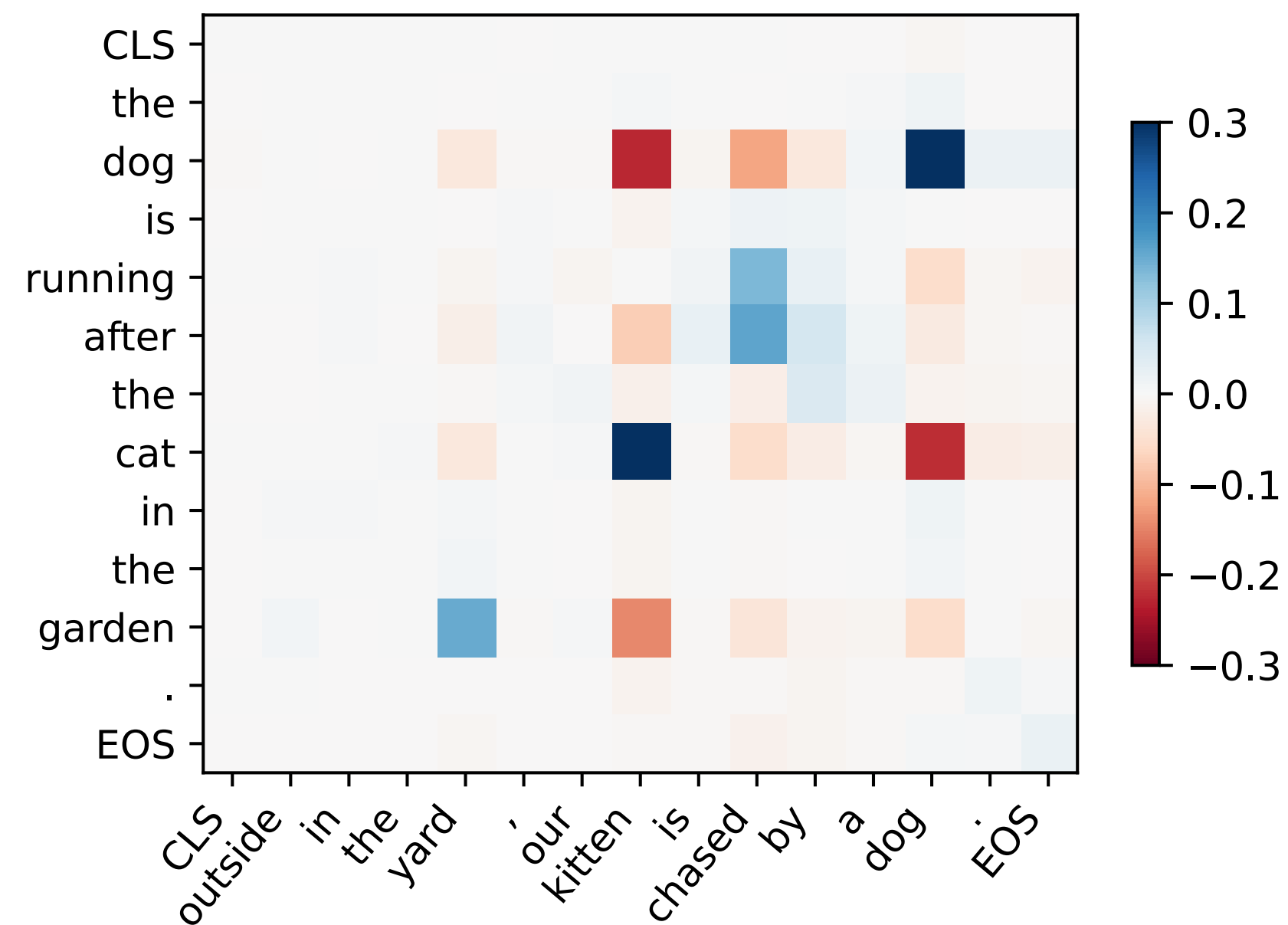
Requires extra modeling and may not perfectly reflect the original model's reasoning. Offers a transparent pathway to analyze otherwise opaque embeddings.



Attribution via Integrated Jacobians

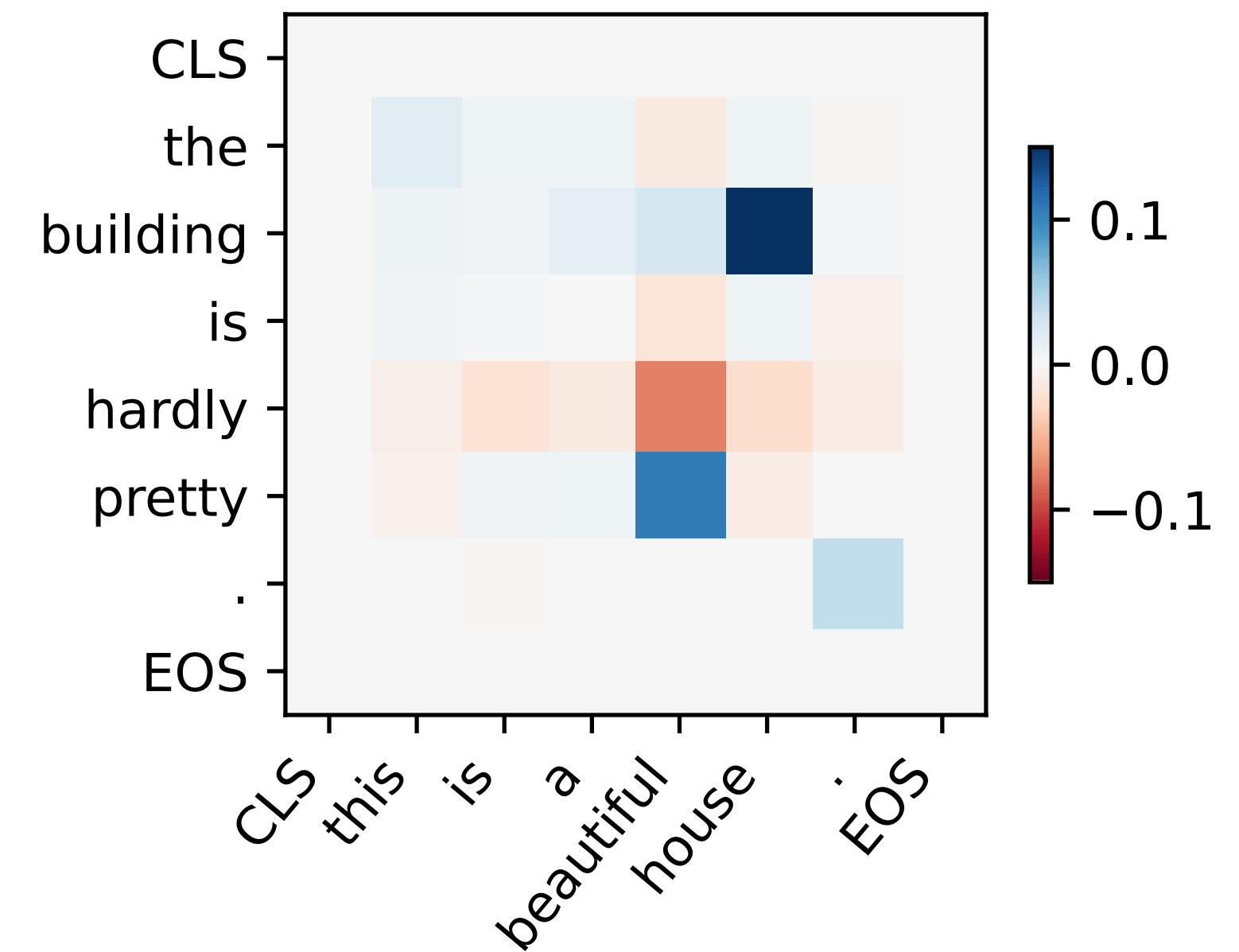
Input A: The dog is running after the cat in the garden.

Input B: Outside in the yard, our kitten is chased by a dog.



Input A: The building is hardly pretty.

Input B: This is a beautiful house.



[1] Moeller et al, An Attribution Method for Siamese Encoders, EMNLP'23

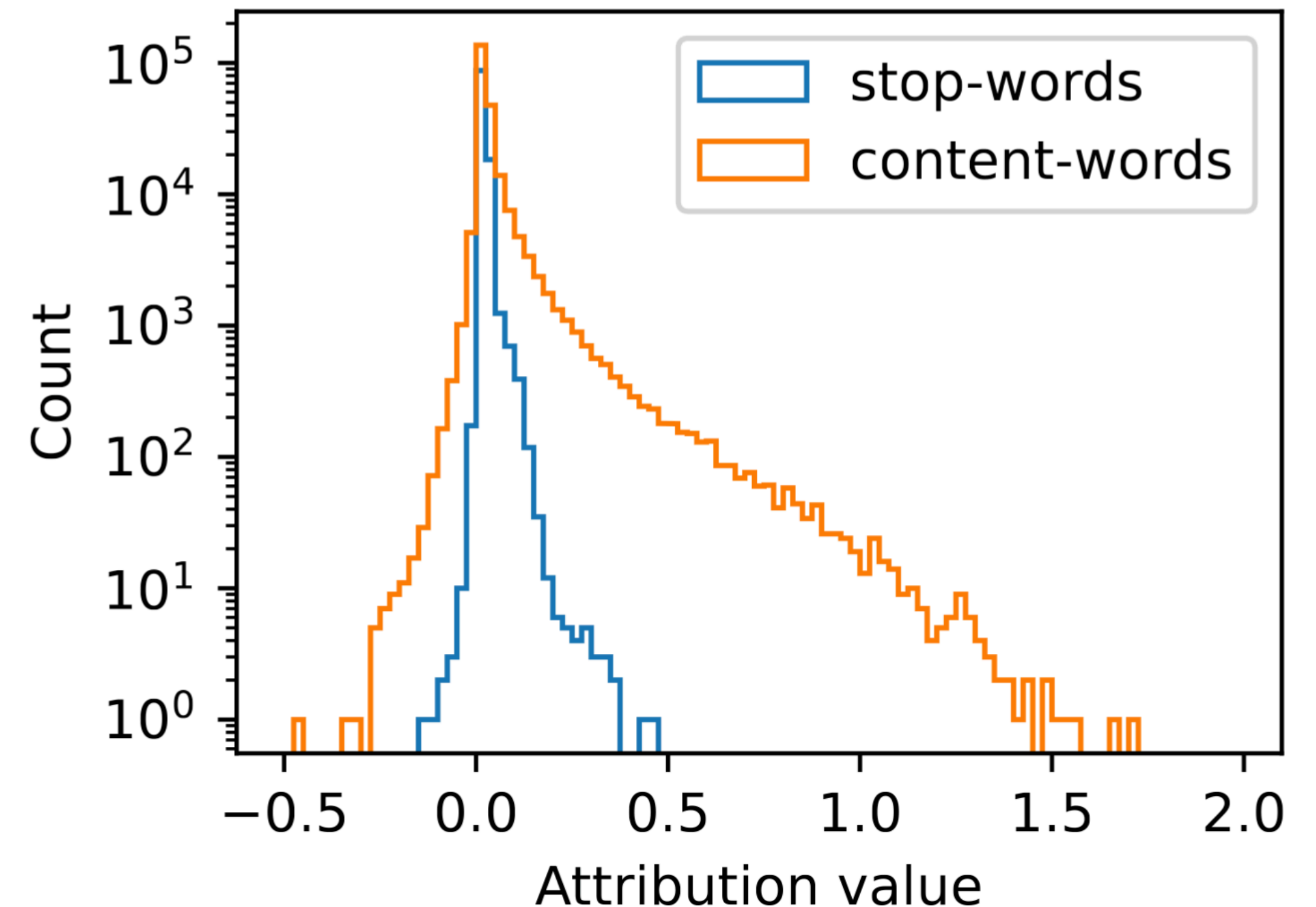
[2] Moeller et al, Approximate Attributions for off-the-shelf Siamese Transformers, EACL'24

github.com/lucasmlr/xsbert

PoC for IR on Stop-words

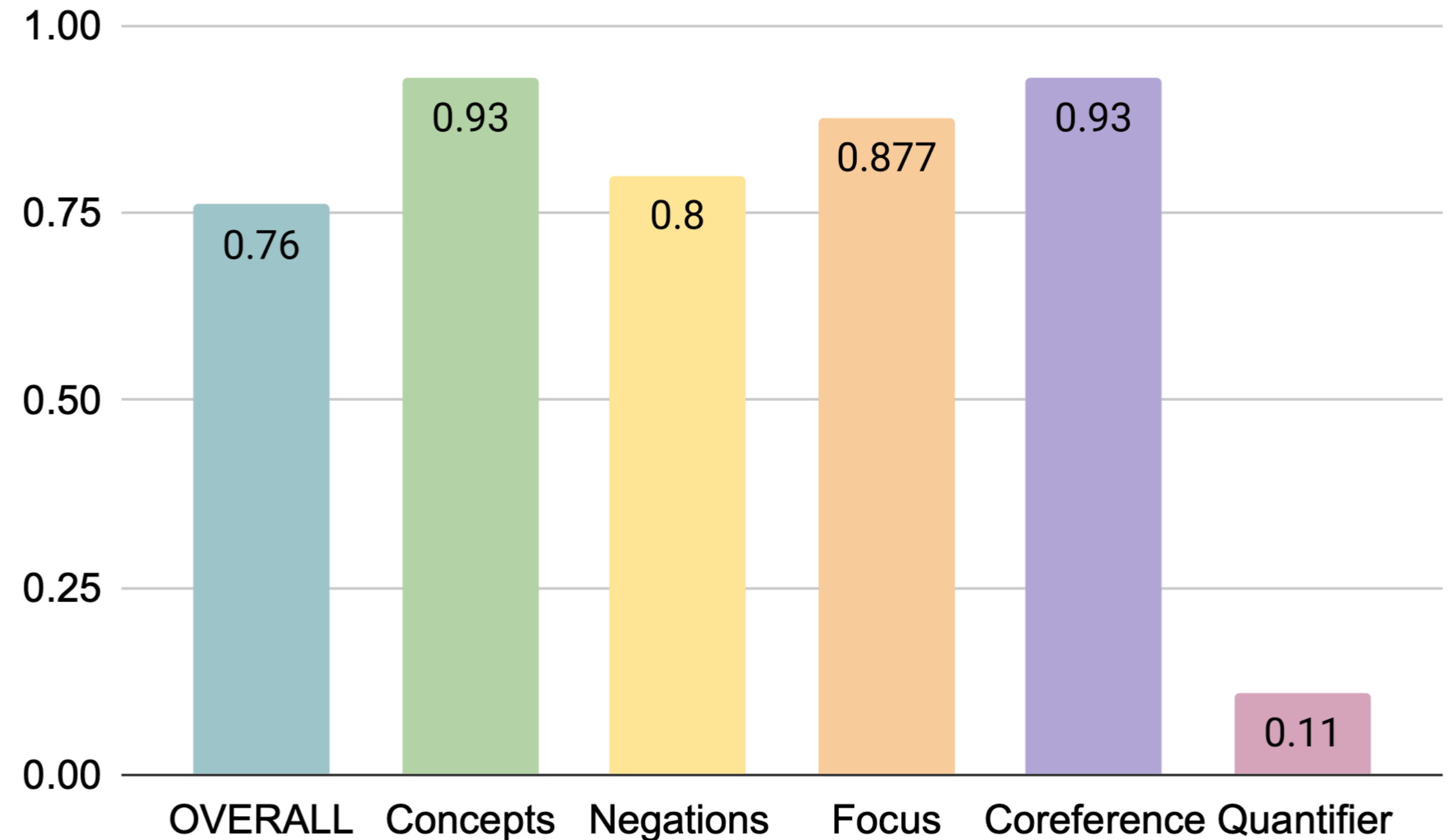
Take-away:

Dense **retrieval models** learn to **ignore stop-words**
and mostly **look at content words**



Space-Shaping via S3BERT

Input A: Two men are singing.
Input B: Three men are singing.



[1] Opitz et al, SBERT studies Meaning Representations:
Decomposing Sentence Embeddings into Explainable Semantic Features
github.com/flipz357/S3BERT

Thank you!

Survey

